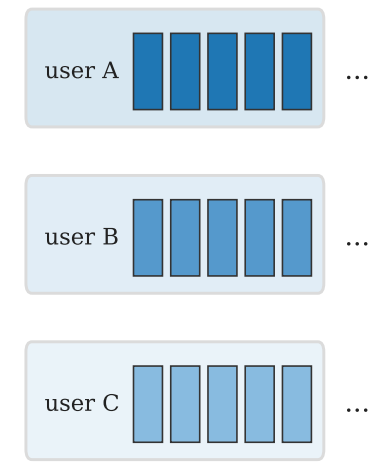
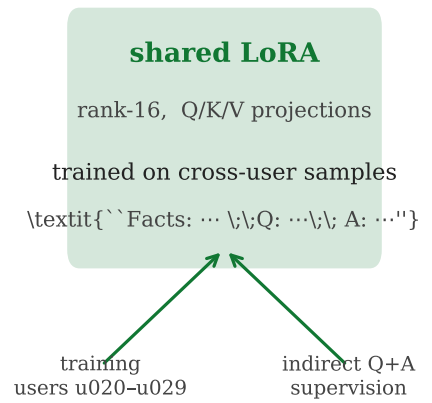


(a) Per-user content
(Engram override, 88\,KB/user)



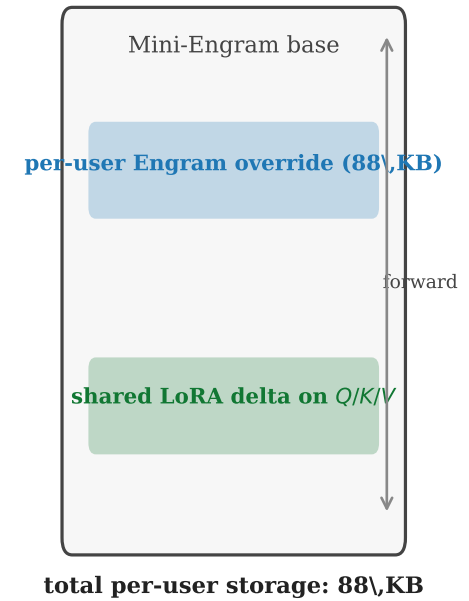
$\Delta\text{bpb on unrelated text} = +0.0001$
(fires only on trigger N-gram)

(b) Shared meta-skill
(one LoRA, 12\,MB, amortised)



$\Delta\text{bpb on unrelated text} = +0.39$
(amortised across 1 M+ users)

(c) At inference
(stacked, both layers live)



(d) Pareto on Mini-Engram-d20
(same base, six substrate combos)

